

Redirecting Client DNS Requests

To restrict client DNS to only the DNS Resolver or Forwarder on pfSense® software, use a port forward to capture all client DNS requests.

❗ Note

Either The DNS Resolver or DNS Forwarder must be active and it must bind to and answer queries on *Localhost*, or *All* interfaces.

❗ See also

- [Blocking External Client DNS Queries](#)
- [Blocking Web Sites Using DNS](#)

The following example uses the LAN interface but the same technique will work with any local interface.

- Navigate to **Firewall > NAT, Port Forward** tab
- Click  **Add** to create a new rule
- Fill in the following fields on the port forward rule:

Interface:

LAN

Protocol:

TCP/UDP

Destination:

Invert Match checked, LAN Address

Destination Port Range:

DNS (53)

Redirect Target IP:

127.0.0.1

Redirect Target Port:

Redirect DNS

NAT Reflection:

Disable

When complete, the port forward must appear as follows:

No RDR (NOT)	☐ Disable redirection for traffic matching this rule This option is rarely needed. Don't use this without thorough knowledge of the implications.		
Interface	LAN Choose which interface this rule applies to. In most cases "WAN" is specified.		
Protocol	TCP/UDP Choose which protocol this rule should match. In most cases "TCP" is specified.		
Source	Display Advanced		
Destination	☒ Invert match. LAN address / Type Address/mask		
Destination port range	DNS Custom DNS Custom From port To port Specify the port or port range for the destination of the packet for this mapping. The 'to' field may be left empty if only mapping a single port.		
Redirect target IP	127.0.0.1 Enter the internal IP address of the server on which to map the ports. e.g.: 192.168.1.12		
Redirect target port	DNS Custom Port Specify the port on the machine with the IP address entered above. In case of a port range, specify the beginning port of the range (the end port will be calculated automatically). This is usually identical to the "From port" above.		
Description	Redirect DNS A description may be entered here for administrative reference (not parsed).		
No XMLRPC Sync	☐ Do not automatically sync to other CARP members This prevents the rule on Master from automatically syncing to other CARP members. This does NOT prevent the rule from being overwritten on Slave.		
NAT reflection	Disable		

Note

If DNS requests to other DNS servers are blocked, such as by following [Blocking External Client DNS Queries](#), ensure the rule to pass DNS to `127.0.0.1` is above any rule that blocks DNS.

With this port forward in place, DNS requests from local clients to **any** external IP address will result in the query being answered by the firewall itself. Access to other DNS servers on port 53 is impossible.

Tip

This can be adapted to allow access to only a specific set of DNS servers by changing the Destination network from "LAN Address" to an alias containing the allowed DNS servers. The **Invert match** box should remain checked.

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Clients using DNS over TLS or DNS over HTTPS could circumvent this protection.

Redirecting or blocking port `853` may help with DNS over TLS, depending on the clients.

See [Blocking External Client DNS Queries](#) for additional advice.